

Security Experiment Worksheet

Grand Marina Hotel — TLS Security Assessment

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Experiment 1: Eavesdropper Test

Question: Can someone spy on our sensor data?

Scenario	What You Observed
Without TLS (Part A)	<i>Example: "Saw all 5 messages with device IDs and pressure readings in plain text"</i>
With TLS (Part B)	<i>Example: "Error: A TLS error occurred. Could not see any messages."</i>

Conclusion (Did TLS block eavesdropping?):

Screenshot:

[Paste your screenshot here]

Experiment 2: Certificate Test

Question: What happens when certificates are wrong?

Scenario	Result	Error Message (if any)
A: Correct certificates	SUCCESS / FAILED	
B: Wrong CA	SUCCESS / FAILED	
C: No verification	SUCCESS / FAILED	

Conclusion (What does this tell you about certificate validation?):

Screenshot:

[Paste your screenshot here]

Experiment 3: Speed Test

Question: Does encryption slow things down?

Metric	TLS OFF	TLS ON
Average latency		
Min latency		
Max latency		

Calculated TLS overhead: _____%

Formula: TLS Overhead = (TLS_avg - NoTLS_avg) / NoTLS_avg x 100%

Conclusion (Is this overhead acceptable for Grand Marina's sensors?):

Screenshot:

[Paste your screenshot here]

Experiment 4: Stress Test

Question: Can the system handle Grand Marina's load?

Rate	TLS OFF	TLS ON
10 msg/sec (Normal)	SUCCESS / DEGRADED	SUCCESS / DEGRADED
25 msg/sec (Moderate)	SUCCESS / DEGRADED	SUCCESS / DEGRADED
50 msg/sec (Emergency)	SUCCESS / DEGRADED	SUCCESS / DEGRADED
100 msg/sec (Beyond req.)	SUCCESS / DEGRADED	SUCCESS / DEGRADED

Conclusion (Can the TLS-secured system handle Grand Marina's load requirements?):

Screenshot:

[Paste your screenshot here]